Physics - Undergraduate Level Assessment 16 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately
   (a) 0.1 GeV/c^2
   (b) 0.2 GeV/c^2
   (c) 0.5 GeV/c^2
   (d) 1.0 GeV/c^2
2. The mean kinetic energy of the conduction electrons in metals is ordinarily much higher than kT because
   (a) electrons have many more degrees of freedom than atoms do
   (b) the electrons and the lattice are not in thermal equilibrium
   (c) the electrons form a degenerate Fermi gas
   (d) electrons in metals are highly relativistic
3. The speed of light inside of a nonmagnetic dielectric material with a dielectric constant of 4.0 is
   (a) 1.2 x 10^9 m/s
   (b) 3.0 x 10^8 m/s
   (c) 1.5 x 10^8 m/s
   (d) 1.0 x 10^8 m/s
4. A refracting telescope consists of two converging lenses separated by 100 cm. The eye-piece lens has a focal length of 20 cm. The angular magnification of the telescope is
   (a) 4
   (b) 5
   (c) 6
   (d) 20
5. A satellite of mass m orbits a planet of mass M in a circular orbit of radius R. The time required for one revolution is
   (a) independent of M
   (b) proportional to $m^(1/2)$
   (c) linear in R
   (d) proportional to $R^(3/2)$

True / False (10 marks)

Answer True or False

6. True or False: During a reversible thermodynamic process, the internal energy of the system changes significantly.
7. True or False: An observer moving at 0.50 c will measure the rod's length as 0.80 m, assuming no relativistic effects.
8. True or False: The primary source of the Sun's energy is thermonuclear reactions involving two hydrogen atoms and one helium atom.
9. True or False: The electric displacement current through a surface S is proportional to the rate of change of the magnetic flux through S.
10. True or False: A photon detector with a quantum efficiency of 0.1 will detect exactly 10 photons out of 100 sent into it.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of the universe's temperature on the distance of galaxies, specifically analyzing the relationship between a temperature of 12 K and their current distances.
12. Discuss how the Hall coefficient can be used to determine the sign of charge carriers in a doped semiconductor, including the underlying principles and implications of your findings.

End of Paper